General Logic

§§ 26–34 (printed pp. 91–105)

The close of the treatment of definitions, and

Fourth Division: On Divisions

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[…] and sometimes also its necessity—as that by substance one must understand the ultimate inward and real ground of all the constitutive properties in an object: *and partly that the criteria adduced occur not only here, but likewise in all the cases designated by a certain word. The latter is proved in turn either by induction from several examples or by deduction,* when from a generally admitted criterion one derives that which, in a real definition, one would rather lay down as the ground. For example, from the fact that virtue is by ordinary usage ascribed neither to God, nor to angels, nor to the saints, it may be concluded that virtue is moral perfection in struggle with sensibility and a refractory nature.

26.

Word-explanations.

Word-explanations—not the same as nominal definitions, § 22—are those which elucidate a word's signification either by other synonymous words or by the word's composition and origin: for example, those of philosophy or of ontology. *The aim of this is partly to prove the agreement of the latter with the word's original and customary meaning.* The latter is especially feasible with derived and compound words. For since those who first formed them presumably conformed therein to the natural kinship they thought they discovered between the signification of the root-words and the new concept they wished to express, one may readily infer from the former what they had in mind with the compound and derived ones. Etymology accordingly serves to find out what words properly signified in the older language, and what they may signify according to their origin. Root-words, on the other hand, one cannot explain otherwise than by showing, through comparison with the derived ones, what the former in all likelihood signified, in case this is now doubtful or forgotten. For that either the whole word, or each letter and syllable, should have its own meaning is scarcely possible to prove. Besides, this sort of explanation serves to show the human understanding's own progress in the formation of certain concepts—for example, of fate, of God, and of spiritual beings—which is sometimes very instructive. Of less importance, to be sure, is the etymology of such words as designate no scientific objects, but only such as occur in daily and civil life: temples, implements, officers of authority, and so on. Nevertheless these serve both to explain many passages in the writers, and to acquaint us more closely with the spirit and customs of antiquity, which can be worth the trouble of knowing.

27.

Which words one ought to explain.

Where definitions or word-explanations ought to be placed, everyone must judge for himself according to the purpose of his exposition. *In general all principal concepts*—that is, those on which doctrines and rules are subsequently grounded—*ought to be explained exactly*, since the least negligence here often leads to incorrect inferences. For every definition contains a proposition that is laid down as the ground of all the considerations one can raise about the object. It is therefore necessary that this foundation be well laid, if all that one builds upon it is not to fall. Indistinct concepts—though they may for the rest be both clear and correct, as those of virtue, duty and right—do not put us in a position to answer every question whose resolution social life and each man's own peace of mind may demand; even though those who study jurisprudence are at first apt to regard definitions of such well-known things as superfluous, and almost everyone imagines he knows in advance what he ought to understand by these words. It is the same with time, space, and other things about whose true essence the philosophers are in doubt, while nothing seems easier for the ignorant to explain.

On the other hand, this diligence can also be overdone, when to no purpose one fills textbooks with definitions of settled concepts or superfluous technical terms, while treating the most important ones cursorily—indeed, sometimes passing them by in complete silence. No science has been more burdened with this kind of superfluity than metaphysics or logic itself, whence it has been carried into other sciences as well. Good taste has sought to purge them of it. Only, what a pity that along with such verbiage it has sometimes swept out much that was good and solid, which it failed to notice amid the rubbish.

28.

Descriptions.

Descriptions are properly intended for explanations of singular objects, whose essential and abiding criteria—as with singular things generally—either are lacking or are entirely unknown. One can therefore adduce none but inessential and changeable ones. No inorganic body seems to have true individuality, since no part of it is abiding, and the form itself can vary in countless ways without one's being able to fix any point at which it ceases to be the same—for example, Noah's Ark or Theseus's ship. In towns and buildings all the criteria, down to the situation itself, are impermanent, as in the case of Tyre. Human beings and the more perfect animals, at least, are beyond doubt real individuals, which through all life's vicissitudes retain, outwardly and inwardly, a certain indelible character; but who is in a position to determine it? We are therefore compelled merely to describe them by name, station, age, deeds, descent, and so forth—although all this may perish while the singular human being or animal still remains the same. One is indeed accustomed, especially where human beings are concerned, to add some traits of character; but these must always remain partly uncertain, partly indeterminate, the essential ones being easier to give occasion to divine, by indistinct hints, than to explain exactly.

In describing general concepts, or genera and species, the investigators of nature adduce, besides the essential or artificial criterion, also all sorts of changeable and accidental ones. For since accidental constitutive properties have, in part, their ground in something abiding and essential, the former may even serve toward a more thorough knowledge of the very essence of things. Besides, they strike the eye soonest, and are therefore easier to find out.

Fourth Division.

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29.

Higher and lower concepts.

A higher concept is a general one under which others, called lower, can be thought as comprehended—as human beings under animal, triangles under figure. A concept that contains only such criteria as several things resemble one another in, while itself being comprehended under a higher one, is called, in relation to the latter, a *species*, and the higher one a *genus*. The same general concept can therefore, in different relations, be regarded both as genus and as species. The highest genus and the lowest species alone form an exception to this; for the former comprehends all, without itself standing under any higher concept, and consequently can never be regarded as a species, while the latter comprehends only singular things, and can therefore not be regarded as a genus.

To the higher concepts one ascribes greater extension—extensio—or range, because they comprehend more singulars; *to the lower, more content or matter*—comprehensio—since, besides the common criteria, they have many peculiar ones.

Concepts are to one another either subordinate or co-ordinate. If the first is the case, as between genera and species, the former are called subordinating, the latter properly subordinate. If, on the other hand, the second, then they are either both comprehended in the same genus, as nearest species—for example, marsh violets and pansies—or belong to different genera—for example, the ones just named and night-violets, i.e. disparate ones. *Species can be subordinate to their genera either mediately or immediately:* in the latter case these are called their *nearest*, in the former their *remoter genera.* Philosophy allows itself many leaps here—for example, from human being to animal—which in natural history would be unpardonable.

The highest genus and the lowest species are either the *absolutely highest and lowest*—such as thing or being; human being, horse, dog, and so forth—or *relative* to this or that science. Thus body is in natural science the highest genus, in metaphysics a much lower one. In anthropology, where the different human races, nations, sexes, ages and so forth are treated, the concept of a human being is the highest, which in general philosophy is the very lowest.

30.

Proper and improper genera or species.

Genera as well as species are partly proper, partly improper or analogical. The criteria of the former are essential and abiding, those of the latter merely accidental and impermanent.

Although both in philosophy and in other sciences one prefers to discover the first kind, because it is most incumbent upon us to know the essential properties of things, *yet the latter kind too are necessary*, when either no other is available (§§ 21 and 23), or when they can be of use in other respects, chiefly in the application of general theories. Thus for prudence in dealing with human beings it is not enough to know human nature in general: the different inclinations and circumstances by which people let themselves be governed must likewise be known to us. Now it is nevertheless impossible to study out the temperament or character of every singular person; and consequently the learned have done us a service in dividing them, according to temperament, sex, age, descent and so forth, into several improper species, whereby the most important differences that can obtain among singular persons are at least brought to notice. For the same reason there are entered in natural history not only all the species regarded as abiding, but even many varieties—for example, of pot-herbs, garden flowers and other natural products—which can be noteworthy both in a theoretical respect and still more for the sake of use. Many higher genera, classes and orders are themselves improper: whence the so-called *artificial systems*, as opposed to the natural ones, have their origin.

31.

A whole and its parts.

A whole is a unity which, together with several things called the parts, taken all together constitutes the same: for example, the world with all the heavenly bodies and what these contain; the state with all its citizens; the genus with all the species belonging to it. These parts may now either lie within and outside one another in such a way that, by virtue of an actual or real connection in the object itself, they must be regarded as inseparable—as in the world, and especially in organic bodies, in which not only does each part have some influence on the rest, but no one of them can so much as subsist by itself—or this relation may exist only in our representations. Thus there is not among all animals of one species any actual community that makes the one dependent on the other. Nevertheless every species constitutes a whole. In the first case they are *real*, in the second *ideal parts*. Singular beings are parts of the species, these of the genera, and all taken together parts of the higher genus—but only in thought. *This difference is very important.* One has often confused the real totality, which is the world, with the ideal one, which is thing or being in general; and both again with the highest unity in principles, or the ultimate cause of all things, on which they depend as effects. Pantheism and the emanation-system rest chiefly upon this misunderstanding. The highest being is neither a general concept, under which spirit and body are comprehended as species, nor yet any whole, like the world, of which all individuals are parts. But in order to root out such representations it is necessary to develop these concepts exactly, though in themselves they seem clear enough.

32.

Logical and other divisions.

From this it follows that a whole can also be divided in two ways: namely into real parts, which actually cohere or are inwardly connected—a resolution into such may be called *partition*—partitio—and into ideal ones, when a genus is resolved into its species, which is properly, or *logical division*—divisio. In the first way one divides countries into their provinces, the body into head, trunk and limbs, civil society into estates, and so on. In the second, natural history and other sciences produce their classes, orders, genera, species and varieties; of which many occur in logic too—for example, in the division of concepts.

Distinction—distinctio—is, by contrast, only a remark upon the difference of several closely related concepts, which do not even always belong to the same genus or species: for example, synonyms, such as luxury and extravagance; sensible, shrewd, rational, and the like. How much this exactness conduces to distinctness in concepts is clear from the foregoing.

The members or limbs of a division may be two, three or more—dichotomy, trichotomy, and so on. *—How many there ought to be is partly already determined by experience, through the essential difference of the natural genera and species*, as in natural history, where sometimes scores of species are given under the same genus; *and partly it must be determined by ourselves according to certain fundamental principles.* But among these, some may favour dichotomy, others trichotomy. Thus the principle that every thing has its opposite, and that all things are thereby both born and perish, leads to dichotomy; whereas the principle that such opposition must always be united by some third, without which no thing could be born or subsist, leads to division into three—which the critical and most recent philosophy of nature therefore prefer. The sacredness or higher significance of the number three is thus acknowledged here as well.

Divisions may be made in several respects, or with regard to several of the whole's inward or outward criteria, *and these are then called the ground of the division.* Since concepts, for example, have both matter and form, they may in the first respect be divided into simple and compound; in the second, into clear and obscure; again, as regards their origin, into pure and empirical, and so forth. In the same ways human beings may be divided according to sex, age, aptitude, station, and so on, each of which yields a particular ground for new species and subspecies.

The division may be continued by subdivisions, when the species are again resolved into new ones. Thus one divides, for example, clear concepts into distinct and indistinct, the first into thorough and superficial (§ 3); animals and plants first into certain classes and families, then into genera, species and varieties.

33.

The utility of divisions.

Divisions are very useful a) for surveying and ordering the whole collection of knowledge one has acquired. Without divisions it would not only be confused, but difficult to retain, to use and to recall. To this is added the pleasure that unity amid so great a manifoldness, and the enjoyment of our whole riches in a general survey, can afford. *This is why the tabular method has found so many admirers in all the sciences.* It consists in setting forth, in order but briefly, all the divisions, subdivisions and partitions of the principal concept or object, down to the most particular; the doctrines belonging to each concept being added in their place, but without proof. This manner is not only an almost indispensable aid to the memory—chiefly in the empirical sciences and disciplines—but is also useful for discerning the real as well as the ideal connection of things. Nevertheless it can be carried too far even in these, when it is made exclusive, and the memory is thereby enriched more than the higher power of thought is sharpened. In the stricter or proper sciences especially, it then leads easily to superficiality and verbiage: of which the Lullian art affords a remarkable proof.

b) For easing and guiding the work of judgment in the application of general concepts or rules. For this reason the division must be continued down to the lowest species of things. In the practical sciences it is especially necessary to descend as deep as possible, without engaging with merely singular cases, from which an endless prolixity would arise within the science itself. In agriculture and horticulture one must therefore know even the most important varieties of trees, grains and other plants, which general natural history passes over. The statesman's knowledge must not confine itself to any merely general knowledge of forms of government and of states. Every individual of a state is important enough to be considered for itself. For the moralist and the popular teacher it is not enough to know the principal species of virtues and vices—to which the diligence of the scholastic philosophers in this matter mostly confined itself. A general discourse on such subjects commonly remains unfruitful, unless one also knows and understands how to apply it to the many sorts of ways in which virtue and vice are wont to vary according to persons' outward and inward condition.

34.

Rules necessary throughout.

In order to derive this utility from divisions, the following rules are at least to be observed. *a) The parts must be enumerated completely, and the whole which one divides must thereby be exhausted*; for otherwise our knowledge of it becomes not only incomplete, but can even, when one nevertheless imagines the contrary, mislead us into false inferences. As if one were to divide salts into alkaline and acid salts, from which it would follow that common salt and sugar belonged either to one of the parts or to another genus. Were one to say that all virtue is either justice or magnanimity, then both wisdom and truthfulness, with several others, would have to be excluded. Were one likewise to divide animals into those with two feet, four feet, and none, then those with six or more feet, such as insects, would come to stand outside this class.

b) They must be mutually opposed, and not comprehend one another—as reciprocity does in causality, although Kant, under the category of relation, makes two species of these, just as he makes infinite propositions a species of their own, distinct from the affirmative and the negative; of which more hereafter. For the members of the division then coincide, and no real division takes place.

c) No leap must be made from higher to lower species; rather the nearest must be set beside one another: for example, if instead of dividing figures into rectilinear and curvilinear, or regular and irregular, one were at once to divide them into circles, triangles, squares and polygons, one would offend at one stroke both against this rule and against the first. No better is the common division of forms of government into monarchic, democratic, aristocratic and despotic—which even combines the aforesaid faults with the following one.

d) No foreign species ought to be introduced into the division. A rule whose correctness is indeed self-evident, but which is nevertheless often transgressed for want of a distinct concept of the whole—as when either despotism, or what is still worse, ochlocracy, indeed even anarchy, is set beside the regular forms of government.